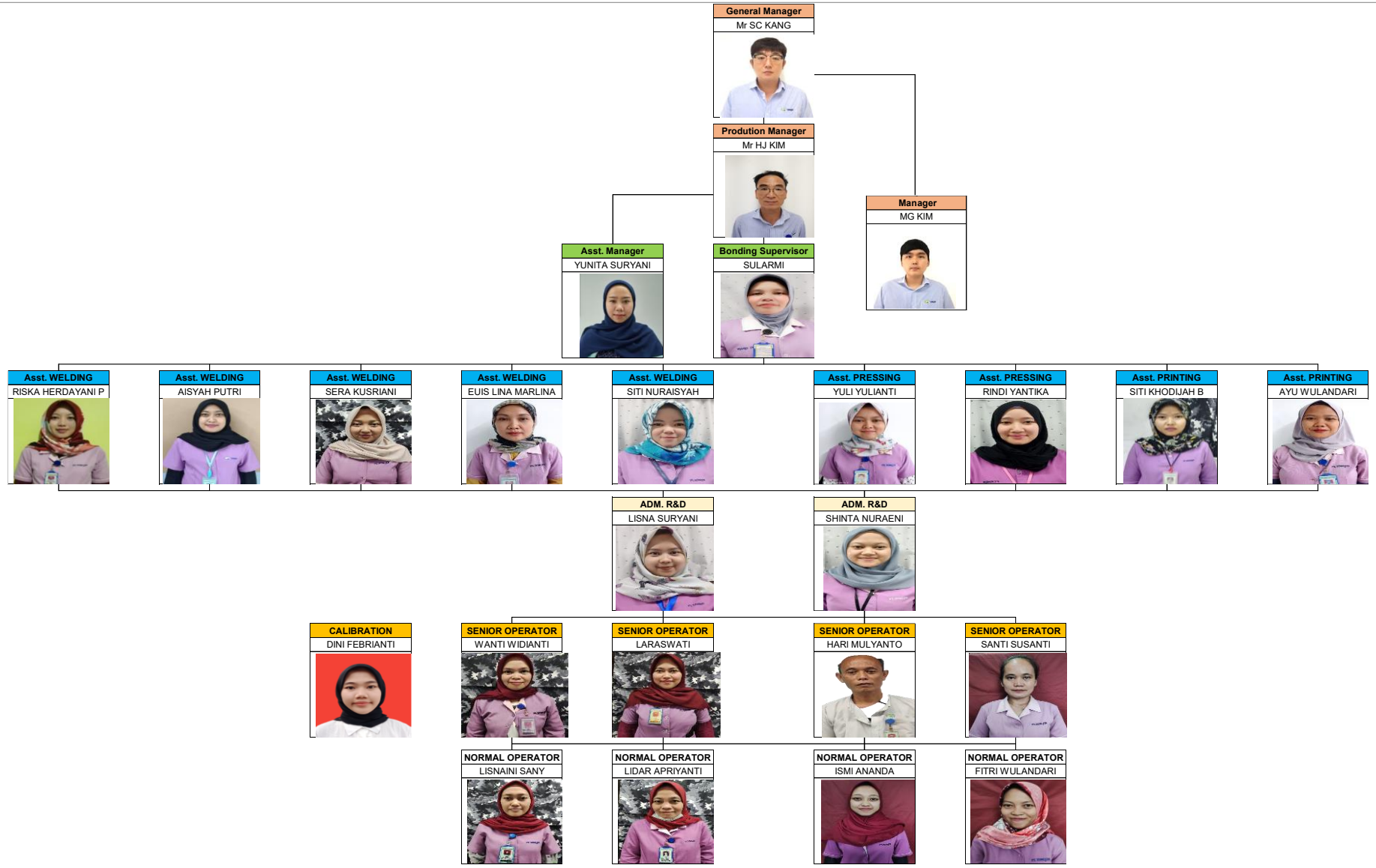
SOP for Welding & Bonding

YONGJIN JAVASUKA GARMENT

MR KIM HANJONG



YJ Bonding Organization Chart



Index

I. Preparation prior to start production

- Check the type of fabric, adhesive, reflective tape and/or heat transfer
- Check Bonding conditions (Temperature,Pressure,Time/speed) & Peel strength/Shear test report conducted by 3rd party lab during development phase
- Appropriate machine selection by work process
- Wash test for quality on mockup
- Test result & Action Plan

II. During Production

- Instruction on working method & Training on how to work
- Hourly machine inspection and calibration report creation and reporting
- Wash test for quality on mockup

III. Inspection and work improvement after production

- Strengthen QC
- Work improvement
- Record keeping

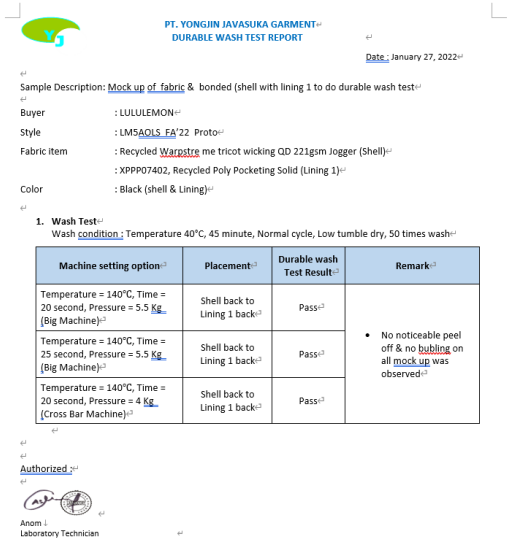
I. Preparation prior to start production

- Check the type of fabric, adhesive, Reflective and/or heat transfer
  - Collect the all of actual quality/color of off-cut fabric and materials received for production
  - Check materials compared to BOM (Tech pack) whether all is matching together correctly
  - Check the approved pre-production sample compared to each items for correct usage/placement on garment

- Check Bonding conditions (Temperature,Pressure,Time/speed) as well as Peel strength/Shear test report
  - Collect all the working conditions (Temperature,Pressure,Time/speed) which resulted in PASS during development phase at 3RD party lab following RMV's recommendation
  - Set the machines following the same conditions of what we've executed tests (peel strength, wash test, shear test) which are PASS at 3RD party lab
  - Make mockups per each set of fabric/color/item in actual for wash test
  - After washing, check the bonded areas if delamination occurred and check for other potential "concern" at earlier stage



Approved Sample at development phase



PT. YONGJIN JAVASUKA GARMENT
DURABLE WASH TEST REPORT

Date : January 27, 2022

Sample Description: Mock up of fabric & bonded (shell with lining 1 to do durable wash test

Buyer : LULULEMON
Style : LM5AQLS_FA'22 Proto
Fabric item : Recycled Warpstre me tricot wicking QD 221gsm Jogger (Shell)
: XPPP07402, Recycled Poly Pocketing Solid (Lining 1)
Color : Black (shell & Lining)

1. **Wash Test**
   Wash condition : Temperature 40°C, 45 minute, Normal cycle, Low tumble dry, 50 times wash

Machine setting option	Placement	Durable wash Test Result	Remark
Temperature = 140°C, Time = 20 second, Pressure = 5.5 Kg (Big Machine)	Shell back to Lining 1 back	Pass	• No noticeable peel off & no bubling on all mock up was observed
Temperature = 140°C, Time = 25 second, Pressure = 5.5 Kg (Big Machine)	Shell back to Lining 1 back	Pass	
Temperature = 140°C, Time = 20 second, Pressure = 4 Kg (Cross Bar Machine)	Shell back to Lining 1 back	Pass	

Authorized ,

Anom
Laboratory Technician

Passed bonding condition at development phase



Machine Setting



US Top load machine

I. Preparation prior to start production

➢ Appropriate Machine Selection for Each Work Process

- Select the suitable machines for each type of Bonding process

  Check the size of the working area and the type of adhesive -> 1st selection by Supervisor -> Production Manager for final confirmation

NO	NAME MACHINE	QTY (UNIT)	PROCESS	PICTURES
1	WAVE FUSHING MACHINE	1 UNIT	Chamber Velcro Cuff Tab Placket Loop Shell-Bemis yard for laser cut Part Bonding Big Size	
2	WELDING & COOLING MACHINE HTM-708	5 UNIT	Chamber Velcro Cuff Tab Placket Loop Shell-Bemis yard for laser cut Part Bonding Big Size	

I. Preparation prior to start production

- Appropriate Machine Selection for Each Work Process

3	WELDING & COOLING MACHINE HTM-709	2 UNIT	Chamber Velcro Cuff Tab Placket Loop Part Bonding Big Size	
4	CROSS OVER PRESS HTM-2116 (CROSS BAR)	24 UNIT	Heat transfer Small panel bonding In line Process Bonding	

I. Preparation prior to start production

➢ Wash test on mockups for Bonding

- Make mockups for each part, color, and machine, fill in the details with a waterproof pen, and conduct a wash test

TYPE	CUT & SEW	BONDED	EMBELLISHMENT / SPECIAL APPLICATION
APPLICATION	X	NEXT TO SKIN DRY SECOND LAYERING → 5OX WASH + 1OX 40°C / Tumble Dry Low OUTERWEAR → 25X Wash 5X Tumble Dry Low	10X WASH +10X DRY 40°C / Tumble Dry Low Re: pleating, laser cutting and perforation – 25x WASH +25X 40°C / Tumble Dry Low
FINISHED PRODUCT	Garment wash: -> 2ND PROTO -> 5X WASH + 5X DRY	Garment wash -> @ SECOND PROTO Wash Care Instructions	Garment wash @ 2ND PROTO +ABRASION PRINTS + ARTWORK ONLY

In accordance with lululemon standard (mockup/garment)

➢ Wash test on mockups for heat transfer

- Based on recent update, we will conduct wash test on mockup up to **15x** cycles

I. Preparation prior to start production

➢ Test result & Action Plan

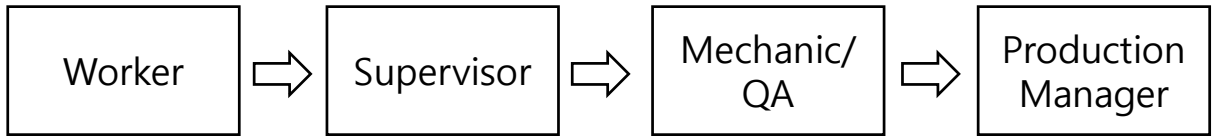
Washing Test FAIL on mockups

- Verify the root cause of problem (machine / fabric / adhesive)
  - > Check the conditions of machine setting
  - > Check the records of machine calibration
  - > Identify the cause/status of problem with operator
- Recalibrate (Temperature,Pressure,Time/speed) where necessary
- Re-wash test

Washing Test PASS on mockups

- All production conditions of mockups test are recorded in detail accurately (Temperature,Pressure,Time/speed)
- After completing the S/O production, it is delivered to the production team after approval by the QA manager

Reporting process in case of problem



II. During Production

- Instruction on working method & Training on how to work
  - Training on how to work with approved S/O standards to follow consistently
  - Strengthen understanding on operator's responsibility and inspecting work quality and Retraining in case of problem occurred
- Hourly machine inspection and calibration report creation and keep recording
  - Check the temperature accordingly and create an inspection log for each device and keep recording on a hourly basis



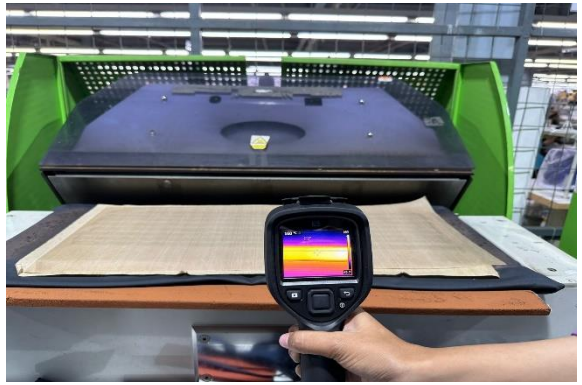
Check the temperature of machine

Pre-test: Heat up the machine 30 minutes before using the machine for process

Method: Measuring the "actual" machine temperature as compared to the "displayed" temperature

Frequency: Every Hour

PASS condition : Should be matching to the machine record



Check the consistency of temperature

Pre-test: Heat up the machine 30 minutes before using the machine for process

Method: Check uniform temperature distribution of the machine

Frequency: Twice a day (before lunch/ after lunch)

PASS condition : Should be within 5 degrees of each other.

II. During Production



Machine Pressure Check

Method: Take a piece of carbon paper or dark colored fabric

Laminate a sheet of glue larger than the bed of the heat press

Frequency: Each Day

PASS condition : No obvious “White Spot patches”

Measures to be taken when machine working conditions are not met

- > Stop processing on the machine in question and report to the manager in charge
- > Re-inspect time for each item when working conditions are not achieved
  (machine temperature: 1 hour before / temperature consistency: before shift / pressure: 1 day before)
- > Speak with the responsible mechanic to check & re-adjust conditions correctly
- > Continue the production if there is no abnormality after the device inspection

When found a problem at work piece

- > Re-inspection of products produced before machine calibration problems
- > Classification of problem items after re-inspection, Fix the processing issue, Conduct appropriate esting
- > Continue the production if there is no problem further

Most of all, keep detailed and organized records to gives the ability to see where problems have occurred and take corrective action when needed immediately.

II. During Production

Daily check list for Machine

PT. YONGJIN JAVASUKA GARMENT	DAILY CHECK LIST CROSS & STOP MARK PRESS MACHINE	No. Dokumen : YJ-F.MEK.0003 Tgl. Efektif : 01 Januari 2017 No. Revisi : 001 Halaman : 1 dari 1

Bulan: Tahun: Line : Mekanik:

DIBUAT	DIPERIKSA	DISETUJUI

NO	TANGGAL	ELECTRICAL POINT				MECHANICAL POINT			
		CHECK POINT	STANDART	HASIL	CORRECTIVE ACTION	CHECK POINT	STANDART	HASIL	CORRECTIVE ACTION
1		HEATER	Berfungsi			BEARING	Tidak bersuara		
		SWICTH HEATER	Berfungsi			SPRING	Tidak rusak		
		POWER SUPPLY	Berfungsi			BOLT	Tidak rusak		
		MAIN BOARD	Berfungsi			UP DOWN (CYLINDER)	Bergerak		
		TEMP DISPLAY	Berfungsi			SPEED CONTROL	Berfungsi		
		SWITCH	Tidak Error			SELANG ANGIN	Tidak bocor		
		SENSOR	Berfungsi			AUTO DRAIN	Berfungsi		
		R/S/T	Berfungsi			TEMPERATURE	± °	± °	

II. During Production

- Wash test for quality on mockups

-> Bonding : Every other day (Considering the time of total wash cycle 25~50x which might take 5~7 days to finish)

-> Heat transfer : Every other day (Considering loss of materials which is not enough to cover massive test)

If any problem on specific date of testing is observed, we will immediately re-inspect all and re-process of it.

Daily check report for Calibration (Temperature)

PT. YONGJIN JAVASUKA GARMENT	Calibration Report (Crossbar)	No. Dokumen : YJ-F.WLD.0003 Tgl. Efektif : 26 Desember 2022 No. Revisi : 00 Halaman : 01

QC	Supervisor	Manager	QA Manager

Date : Machine : Name :

Buyer	Style	S/O Approved Condition			Machine Setting			Hours	Actual					Check	Corrective Action
		Temperature	Pressure (Psi)	Time	Temperature	Pressure (Psi)	Time		Range Target Temperature ±5(°C)			Temperature Consistency	Pressure		
									Left	Center	Right				
								7 - 8	-		-				
								8 - 9	-		-	-	-		
								9 - 10	-		-	-	-		
								10 - 11	-		-	-	-		
								11 - 12	-		-	-	-		
								13 - 14	-		-		-		
								14 - 15	-		-	-	-		
								15 - 16	-		-	-	-		
								16 - 17	-		-	-	-		
								17 - 18	-		-	-	-		
								19 - 20	-		-	-	-		
								20 - 21	-		-	-	-		

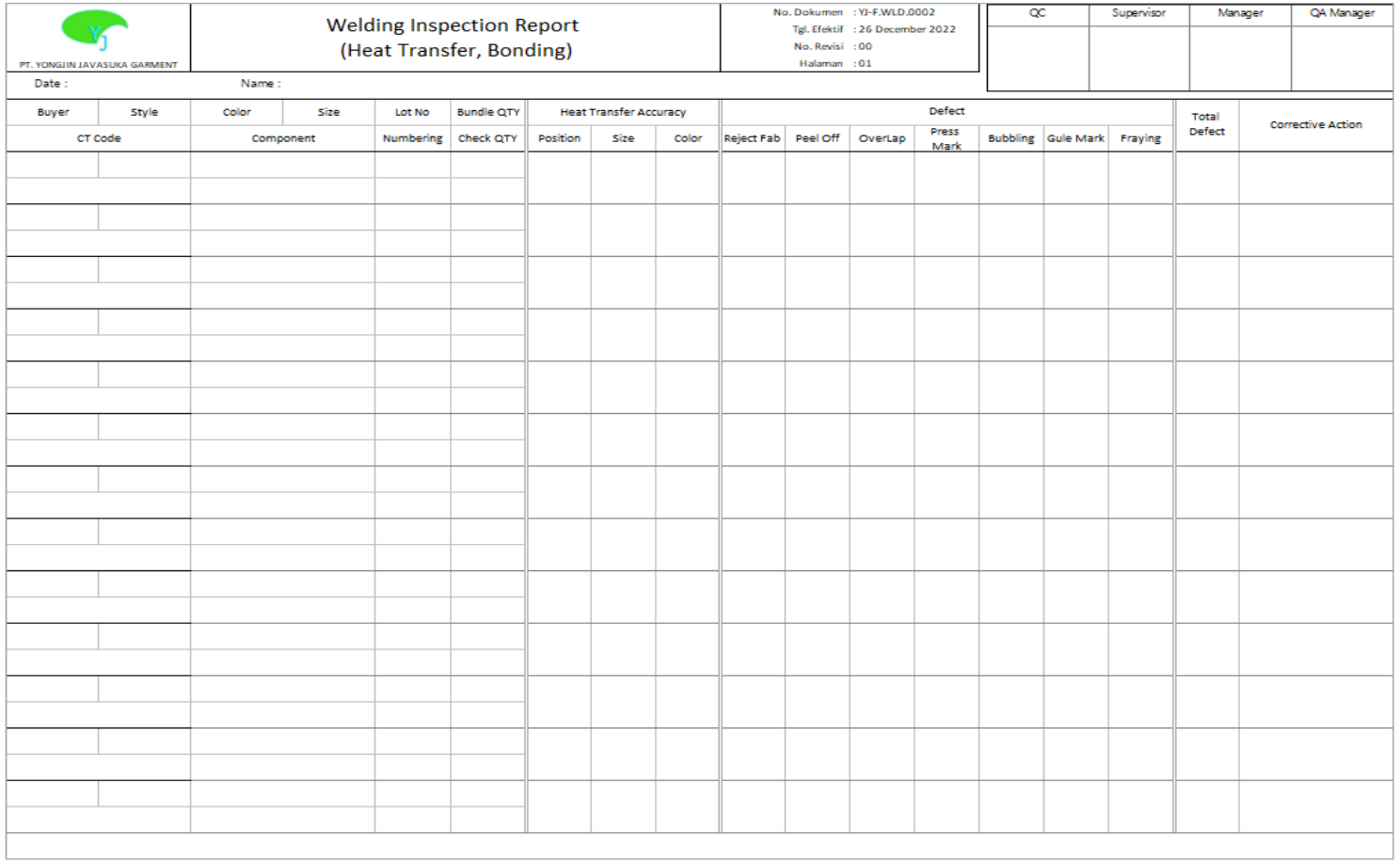
Person in Charge :	Supervisor :	Mechanic :	Manager :

III. Inspection and work improvement after production

- Strengthen QC
  - Recruitment of QC manpower (From 10 -> To 15)
  - QC Inspection Report –Daily/weekly collection of main defect status by style and intensive management (Report to Supervisor, QA, and manager)
  - Add required inspection Defects

Visible Defects

- Press mark (shiny or glaze mark)
- Overlap
- Peel off (Un-bond step down layer)
- Bubbling
- Fraying
- Glue mark (stain, exposed glue)
- Fabric Defect
  - ① Fabric color change from heat
  - ② Bleed through
  - ③ Harsh hand feel
  - ④ Open edge/sharp point
  - ⑤ Crease line/ scar at decorative film



PT. YONGJIN JAVASUKA GARMENT

Welding Inspection Report
(Heat Transfer, Bonding)

No. Dokumen : YJ-F.WLD.0002
Tgl. Efektif : 26 December 2022
No. Revisi : 00
Halaman : 01

QC	Supervisor	Manager	QA Manager

Date :

Name :

Buyer	Style	Color	Size	Lot No	Bundle QTY	Heat Transfer Accuracy			Defect							Total Defect	Corrective Action
CT Code		Component		Numbering	Check QTY	Position	Size	Color	Reject Fab	Peel Off	OverLap	Press Mark	Bubbling	Gule Mark	Fraying		

III. Inspection and work improvement after production

➢ Work improvement

- Improvement of reporting system (Detailed reporting system for each case)
- Composing Calibration team

➢ Record

- Log all tests to prevent reoccurrence of specific problems
- Record the conditions of each machine (temperature, pressure, speed/time) for reference
- Keep washing test results
- Manager reviews records regularly